



Fracture of bicrystal metal/ceramic interfaces: A study via the mechanism-based strain gradient crystal plasticity theory



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Abstract

Two continuum mechanical models of crystal plasticity theory namely, conventional crystal plasticity theory and mechanism-based crystal plasticity theory, are used to perform a comparative study of stresses that are reached at and ahead of the crack tip of a bicrystal niobium/alumina specimen. Finite element analyses are done for a stationary crack tip and growing cracks using a cohesive modelling approach. Using mechanism-based strain gradient crystal plasticity theory the stresses reached ahead of the crack tip are found to be two times larger than the stresses obtained from conventional crystal plasticity theory. Results also show that strain gradient effects strongly depend on the intrinsic material length to the size of plastic zone ratio (l/R_0). It is found that the larger the (l/R_0) ratio, the higher the stresses reached using mechanism-based strain gradient crystal plasticity theory. An

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insight into the role of cohesive strength and work of adhesion in macroscopic fracture is also presented which can be used by experimentalists to design better bimetals by varying cohesive strength and work of adhesion.



Keywords

Metal/ceramic interface; Mechanism-based strain gradient crystal plasticity; Cohesive model; Macro/micro fracture analysis

1. Introduction

Metal/ceramic interfaces play a vital role in modern materials technology, as evident by their use in a variety of applications. For example, high-strength materials such as metal–matrix composites consist of internal interfaces between ceramic (e.g. SiC or Al₂O₃) particles or filaments within a metallic host. Because of their high melting points, extreme hardness, and favourable corrosion resistance, ceramic coatings are often used to protect an underlying metallic component in environments subject to high temperatures and wear rates (i.e., cutting tools). In microelectronics packaging, interfaces between metallic (Cu and/or Al) interconnects and SiO₂, carbide/nitride (TiCN) or oxide (Al₂O₃) ceramics are commonplace, and impact the performance and longevity of solid state devices. Despite their widespread use, a basic understanding of these interfaces has been elusive. For example, given a particular metal/ceramic interface, it is not yet possible to accurately predict such fundamental properties as its fracture energy. In most of the cases, improvements in interface properties proceed via a costly and time consuming trial-and-error process in which numerous materials are evaluated until suitable performance is obtained. Computational methods provide a wide range of possibilities to study the fracture behaviour of such metal/ceramic interfaces. Scientists and engineers have been studying different metal/ceramic interfaces for a long time, in order to create a better understanding of the factors that influence the fracture behaviour of the metal/ceramic interfaces, such as plasticity induced around the crack tip and interface strength.

In order to study and understand the adhesion at a fundamental quantum-mechanical level (Methfessel et al., 1992, Finnis et al., 1999, Zhang and Smith, 2000) ab initio calculations have been performed. Methfessel et al. (1992) performed

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ab initio calculations to compute the surface energies of 4d transition metals (Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag). It was found that surface energies change with changing orientation of single crystalline materials, for example in the case of niobium, surface (100) is found to have highest surface energy as compared to surfaces (110) and (111).

[Finnis et al. \(1995\)](#) performed ab initio calculations of metal/ceramic interfaces. These calculations not only predicted the electronic structure of the interface but also the ideal work of adhesion. The calculations were based on niobium(111)/ α -alumina(0001) interfaces. The computed value of ideal work of adhesion of a niobium monolayer to the oxide surface amounts to 9.8J/m^2 . [Baitrev et al. \(1999\)](#) also performed the ab initio calculations for niobium/alumina interfaces. The work of adhesion for O- and Al-terminated niobium(111)/ α -alumina(0001) interfaces was reported to be in the range of $9.8\text{--}10.8\text{J/m}^2$. It was found that after relaxing the atoms, alumina is always found to oxygen terminated. It was also found that oxygen terminated alumina at the interface has higher work of adhesion ($\sim 9.8\text{J/m}^2$) as compared to aluminium terminated alumina ($\sim 2.8\text{J/m}^2$). [Zhang and Smith \(2000\)](#) examined the relative stability of both stoichiometric and non-stoichiometric niobium(111)/ α -alumina(0001) interfaces. The work of adhesion was computed using two different techniques local-density approximation (LDA) (for details see [Zhang and Smith, 2000](#)) and generalized-gradient approximation (GGA) (for details see [Zhang and Smith, 2000](#)). The work of adhesion computed was consistent with the results of [Baitrev et al. \(1999\)](#).

The mechanical properties of single crystalline sapphire/niobium/sapphire joints were investigated ([Soyez, 1996](#), [Soyez et al., 1998](#)) for different orientation relationships between metal and ceramic. The orientation of single crystalline materials was found to have significant effect on the energies of the Nb/ Al_2O_3 interface fracture. The influence of the chemical composition of regions at or close to the interface on the strength of a metal/ceramic interface was studied ([Fischmeister, 1987](#)). It was found that the strength of a metal/ceramic interface depends upon the density of the metal–oxide bonds. Ni/ Al_2O_3 and Cu/ Al_2O_3 joints were studied ([Nicholas and Lee, 1989](#), [Nicholas et al., 1980](#), [Kurkjian and Kingery, 1956](#)), and it was found that higher strength values can result from adding reactive alloying elements, such as, chromium and titanium. This addition of alloying elements raises the work of adhesion.

Various research groups ([Evans et al., 1990](#), [Evans and Dalgleish, 1992](#), [McMahon and Vitek, 1979](#), [Jokl et al., 1980](#), [Hirth and Rice, 1980](#), [Rice et al., 1990](#)) postulated that the fracture energy of interfaces depends not only on the work of adhesion W_{adhesion} , but also on the amount of plasticity that occurs in the metal during the fracture process, on the interfacial flatness, and on their interrelation. [Fuller et al., 1980](#), [Chiao and Clarke, 2003](#) observed dislocation emission from crack tips, and also proposed that tipility of a crack tip can be characterized by its response to mechanical loading. [Fischmeister et al., 1988](#) found

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that the fracture energy of niobium/sapphire and copper/sapphire interfaces change dramatically with the relative crystallographic orientation of the two constituents. [Beltz and Rice, 1992](#), [Beltz and Wang, 1992](#), [Korn, 1993](#), [Korn et al., 2002](#) observed that the amount of plasticity initiated in the metal during fracture of such interfaces is strongly influenced by both, the interface chemistry and the orientations of slip systems relative to the crack surface. [Kysar \(2000\)](#) studied experimentally the copper/sapphire interface and the directional dependence of the crack for various orientations. [Jameel et al. \(2003\)](#) studied the brittleness and ductility of cracks along the interface of copper bicrystals. The brittleness and ductility of the crack tip was explained in terms of angles that various slip systems make with the crack flanks. They found out that the relative angles of slip planes play a critical role in the process of deformation at the crack tip.

Continuum mechanical studies have been done by many researchers to study and understand the fracture behaviour of metal/ceramic interfaces. [Armstrong, 1966](#), [Price and Kelly, 1964](#) proposed that the distinction between the intrinsic brittleness and ductility can be considered as a competition between cleavage failure and plastic shear at the crack tip.

[McHugh et al., 1989](#), [Varias et al., 1990](#) studied aluminium/rigid and copper/rigid material interfaces using rate dependent crystal plasticity theory ([Asaro, 1979](#), [Asaro, 1983a](#), [Asaro, 1983b](#)). A hybrid decohesion model was proposed to describe the crack propagation along the interface. These results show that the interface strength changes the stress–strain field around a crack tip. It was also found that increasing the interface strength increases the J -resistance of the bimaterial system.

[Clegg et al., 1988](#), [Beltz and Rice, 1992](#) used rate dependent crystal plasticity theory, whereas [Asaro, 1983a](#), [Asaro, 1983b](#) studied the crack tip behaviour of a copper/sapphire interface. [Beltz and Rice, 1992](#), [Beltz and Wang, 1992](#) interpreted the experimental results of [Beltz and Wang \(1992\)](#) based on dislocation nucleation arguments. It was shown that brittle failure occurs when the energy required for dislocation nucleation is larger than the energy required for cleavage, while failure is ductile if the energy required for dislocation nucleation is less than the energy required for cleavage.

[Wang and Anderson \(1991\)](#) performed a series of interfacial fracture experiments on specially oriented copper symmetric tilt bicrystals and reported a directional dependence of crack growth; that is the interfacial crack oriented in one crystallographic direction propagates much more readily than the crack on the same interface but oriented to propagate in the opposite crystallographic direction. [Wang \(1998\)](#) proposed a micromechanical model for interface cracking. Three different systems were studied, i.e., gold/sapphire, aluminium/sapphire and nickel/sapphire. The model proposed in their work, predicted the crack tip

behaviour, i.e., ductility or brittleness of the crack tip by including the local mode mixity at the crack tip in the model ([Beltz and Wang, 1992](#)).

In addition to the dislocation nucleation analyses, there have been other approaches to predict relative brittleness and ductility. [Kysar, 2001a](#), [Kysar, 2001b](#), [Kysar and Briant, 2002](#) studied the directional dependence of crack growth at copper/sapphire interfaces. Finite element analyses of copper/sapphire bicrystal specimen were performed using crystal plasticity theory. It was shown that relative ductility/brittleness can be explained on the basis of the normal opening stress.

[Nakatani et al. \(2003\)](#) analyzed the small-scale yielding around a stationary crack along a ductile single crystal/rigid material interface. Both continuum slip and discrete dislocation plasticity theory were used with either two or three slip systems. Results obtained using both theories were found to be consistent. The discrete dislocation simulations provide a valuable window into the actual micromechanics of how near-tip plastic deformation develops, and what the actual underlying dislocation distribution is that corresponds to the macroscopic elastic–plastic deformation pattern. However limitations on how many dislocations can be handled and how many computer runs can be made, make it difficult to determine the near-tip stress and deformation fields that occur on a larger scale near such a crack tip. This information can be obtained from the continuum slip crystal plasticity solutions, which gave full-field results for the elastic–plastic continuum slip crystal plasticity solutions. It was also shown that both, discrete dislocation simulations and continuum slip crystal plasticity theory accurately predict the crack tip opening behaviour, which is important in understanding the type of failure anticipated.

[Kohnle et al., 2000](#), [Kohnle et al., 2002](#) analyzed polycrystalline niobium/sapphire interfaces. The influence of the plastic properties of the metal part on the interface strength and on the energy release rate is examined. It was shown that low yield stress results in high plastic deformation in the metal part and consequently higher energy is stored before the critical stress value for fracture is reached.

Crystal orientation effects on the crack initiation energies of the bicrystal niobium/alumina specimen have been studied ([Siddiq and Schmauder, 2005b](#), [Siddiq and Schmauder, in press](#)). The results show that by changing the orientation of the single crystalline material, one can change the amount of plasticity induced due to the activation of different slips systems, also, the level of stresses induced at and near the crack tip can be altered. The reason of variation in fracture energies for various orientations were explained on the basis of induced stresses and plasticity around the crack tip.

Interface fracture of a metal/ceramic bicrystal specimen using a cohesive modelling approach and conventional crystal plasticity theory was studied by [Siddiq and Schmauder \(2006\)](#). The effect of different cohesive law parameters, such as, cohesive strength and work of adhesion on the macroscopic fracture energy was studied. An insight on the role of the cohesive strength and work of adhesion on fracture energy was presented. It was found that the effect of cohesive strength on fracture energies is more profound as compared to the work of adhesion. A comparison between experimental and simulation results was presented and a theoretical interlink among cohesive strength, work of adhesion, yield stress and macroscopic fracture energy was presented.

Fracture analysis of a bimaterial interface using conventional theory of mechanism-based strain gradient plasticity was performed by [Qu et al. \(2004\)](#). It was shown that by using the conventional theory of mechanism-based strain gradient plasticity theory, the tensile stress at a distance of $0.1\text{ }\mu\text{m}$ to the interface crack tip reaches $13.3\sigma_y$, where σ_y is the yield stress of niobium, which when compared to classical plasticity theory ($3.6\sigma_y$) is nearly 4 times higher ([Qu et al., 2004](#)) which is due to the strong strain gradient effects near the interface.

The aim of the presented work is to perform a comparative study of the tensile stresses at the interface crack tip using mechanism-based strain gradient crystal plasticity theory [Han et al., 2005a](#), [Han et al., 2005b](#) and conventional crystal plasticity theory ([Hill and Rice, 1972](#)). The study will be carried out, both for a stationary crack tip and for growing cracks using a cohesive modelling approach ([Scheider and Brocks, 2003](#)).

2. Conventional crystal plasticity theory

Crystal plasticity theory is based on the assumption that plastic deformation is the sum of the crystalline slip in all activated slip systems. [Schmidt \(1931\)](#) pointed out that plastic slip occurs when the resolved shear stress onto a crystallographic plane in the direction of slip reaches a critical value. The precise theory was formulated by [Hill and Rice \(1972\)](#).

The resolved shear stress $\tau^{(\alpha)}$, on the α th slip system is computed from the stress tensor σ_{ij} as

$$\tau^{(\alpha)} = \sigma_{ij} \mu_{ij}^{(\alpha)} \quad (1)$$

with $\mu_{ij}^{(\alpha)}$ being the Schmidt factor given by

$$\mu_{ij}^{(\alpha)} = \frac{1}{2} \left(s_i^{(\alpha)} n_j^{(\alpha)} + s_j^{(\alpha)} n_i^{(\alpha)} \right) \quad (2)$$

where $\underline{s}^{(\alpha)}$ is the vector tangent to slip system α and $\underline{n}^{(\alpha)}$ is the vector normal to slip system α .

While the plastic strain rate is given by

$$\dot{\epsilon}_{ij}^p = \sum \mu_{ij}^{(\alpha)} \dot{\gamma}^{(\alpha)} \quad (3)$$

where $\dot{\gamma}^{(\alpha)}$ is the shear strain rate in slip system α .

The stress rate is then defined as

$$\dot{\sigma}_{ij} = C_{ijkl} \dot{\epsilon}_{kl}^e = C_{ijkl} (\dot{\epsilon}_{kl} - \dot{\epsilon}_{kl}^p) \quad (4)$$

A viscoplastic constitutive model is used in this study in the form of a power law expression as

$$\dot{\gamma}^{(\alpha)} = \dot{\gamma}_0^{(\alpha)} \operatorname{sgn}(\tau^{(\alpha)}) \left| \frac{\tau^{(\alpha)}}{g^{(\alpha)}} \right|^m \quad (5)$$

where $g^{(\alpha)}$ is the current strength of each slip system α , γ_0 is the reference (initial) shear strain, while n is the strain rate sensitivity parameter (a larger value of n ensures the rate independent case, such as $n=50$). As $n \rightarrow \infty$, the plastic constitutive formulation formally becomes rate-independent.

The rate of change of the current strength ($g^{(\alpha)}$) of each slip system α is given by

$$\dot{g}^{(\alpha)} = \sum_{\beta=1}^N h_{\alpha\beta} \dot{\gamma}^{(\beta)} \quad (6)$$

where the matrix $h_{\alpha\beta}$ contains the hardening moduli for each slip system. $h_{\alpha\alpha}$ is known as self hardening (hardening of a slip system during deformation) while $h_{\alpha\beta}$ ($\alpha \neq \beta$) is known as latent hardening (hardening of a slip system due to the hardening in another slip system).

Different hardening models have been developed since the evolution of crystal plasticity theory (Taylor, 1938, Nakada and Keh, 1966, Peirce et al., 1982, Bassani and Wu, 1991). For the present study Bassani and Wu (1991) has been used to simulate the behaviour of single crystalline niobium. Bassani and Wu (1991) model is based upon the analytical characterization

of the hardening moduli at any of the three stages during deformation. Their expression for self and latent hardening depends on the shear strains γ^α of all slip systems

$$h_{\alpha\alpha} = \left\{ (h_0 - h_s) \operatorname{sech}^2 \left[\frac{(h_0 - h_s) \gamma^\alpha}{\tau_s - \tau_0} \right] + h_s \right\} G(\gamma^\beta; \beta \neq \alpha) \quad (7)$$

$$h_{\alpha\beta} = q h_{\alpha\alpha} (\alpha \neq \beta)$$

$$G(\gamma^\beta; \beta \neq \alpha) = 1 + \sum_{\beta \neq \alpha} f_{\alpha\beta} \tanh(\gamma^\beta / \gamma_0)$$

where, h_0 is the initial hardening modulus, τ_0 the initial yield stress, τ_s the saturation stress, γ^α the total shear strain in slip system α , h_s the hardening modulus during stage I deformation, $f_{\alpha\beta}$ the interaction strength between slip system α and β , and γ^β the total shear strain in slip system β while sech is the secant hyperbolic function.

The hardening moduli are described with an initial hardening (h_0) which saturates after reaching the resolved shear stress (τ_s). After the diminishing of the hyperbolic secant term, the saturation hardening (h_s) term specifies that each slip system has a finite hardening rate. The function G deals implicitly with cross-hardening that occurs between slip systems during stage II hardening.

Other crystal plasticity models based on dislocations mechanics are also available ([Cuitino and Ortiz, 1992](#), [Kameda and Zikry, 1998](#), [Ohashi and Asakawa, 2001](#), [Harder, 1999](#), [Acharya, 2001](#), [Arsenlis and Parks, 2002](#)). These models explicitly account for the quantification of dislocations density evolution and the kinetics of dislocation interaction. These types of models involve a large number of parameters and coefficients that must be derived from experiments. [Cuitino and Ortiz \(1992\)](#) proposed a viscoplastic dislocation mechanics based crystal plasticity model using statistical mechanics. This model requires in total 12 parameters to be identified for each family of slip system from experiments. [Kameda and Zikry \(1998\)](#) proposed a model by including the mobile and immobile dislocation densities as internal variables in the constitutive formulations. The relation for the evolution of mobile and immobile dislocation densities that correspond in an average sense to the generation, interaction, trapping and recovery of dislocations. This model requires the identification of seven parameters for each family of slip system from uniaxial tension/compression experiments. Ohashi (2001) proposed a model in which the evolution of plastic slip was defined using models of dislocation movement for fee crystals. The mean free path in this model is described as a function of the densities of statistically stored as well as geometrically necessary dislocations. This model is scale dependent where the edge and screw components of the geometrically necessary dislocations are obtained from the strain gradients. This model requires eleven

identified for each family of slip system from experiments. [Harder \(1999\)](#) proposed a crystallographic model for

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pure fee metals in the low temperature range. Rate dependent constitutive equations are based on either isotropic as well as kinematic hardening, whereby the mutual interactions of dislocation processes on the different slip systems are taken into account. This model requires the identification of eleven parameters and was applied to fee metals in the past ([Harder, 1999](#)). [Arsenlis and Parks, 2002](#), [Acharya, 2001](#), proposed a continuum constitutive model based on internal variables characterizing crystallographic dislocation densities. The evolution of plastic strain is given by Orowan's relation as a function of dislocation flux and average dislocation velocity. A large number of parameters are to be identified in this model, approx. 6–18, depending on the dislocation interactions. The above discussed dislocations based models leads to the conclusion that these models explicitly account for such things as the quantification of dislocations density and kinetics of dislocation interactions. On the other hand, these types of crystal plasticity models involve many parameters and coefficients that have to be derived from experiments. Also, limitations on how many dislocations can be handled and how many computer runs can be made, make it difficult to determine the near-tip stress and deformation fields present on a larger scale near a crack tip ([Nakatani et al., 2003](#)).

Present work is performed using the crystal plasticity theory ([Hill and Rice, 1972](#)) along with the two hardening laws by [Peirce et al., 1982](#), [Bassani and Wu, 1991](#), respectively, has been implemented numerically within the UMAT written by [Huang \(1991\)](#). The numerical implementation of conventional crystal plasticity theory has been discussed in detail in [Huang, 1991](#), [Kysar, 2001a](#), [Kysar, 2001b](#). This UMAT subroutine has been used with ABAQUS for the present work.

The crystal plasticity model has been implemented for finite deformations and rotations. The Jaumann stress rate is used in the formulation, which is the corotational stress rate on axes that rotate with the crystal lattice.

The orientations of the slip systems are updated during every call to the UMAT subroutine and stresses are updated using the updated orientations of the slip systems.

3. Mechanism-based strain gradient crystal plasticity theory

Steep strain gradients in the vicinity of the crack tip produce locally high levels of strain hardening due to a high density of geometrically necessary dislocations which ultimately end up with higher levels of stresses around the crack tip. Various plasticity models have been proposed ([Huang et al., 2000a](#), [Huang et al., 2000b](#), [Huang et al., 2004](#); [Shi et al., 2000](#), [Guo et al., 2001](#), [Hwang et al., 2002](#), [Wallin and Ristinmaa, 2005](#)) to study the effect of plasticity length scale at micron and submicro scales.

Typesetting math: 100% approach to study the size effects through strain gradients is adding of a variable (tensorial) involving the

gradients of plastic strain. There are some models ([Acharya and Bassani, 2000](#)) where strain gradients are incorporated in the hardening moduli only, this way the classical mathematical framework of standard plasticity is preserved.

Strain gradient crystal plasticity formulations have been proposed by many groups ([Smyshlyaev and Fleck, 1996](#), [Shu and Fleck, 1998](#), [Acharya and Bassani, 2000](#), [Regueiro et al., 2002](#), [Shizawa and Zbib, 1999](#), [Menzel and Steinmann, 2000](#), [Gurtin, 2002](#), [Yefimov and Van der Giessen, 2005](#), [Han et al., 2005a](#), [Han et al., 2005b](#), [Cheong et al., 2005](#), [Ohashi, 2005](#)). These continuum models take into account for the different role of geometrically necessary dislocations and statistically stored dislocations. [Acharya and Bassani \(2000\)](#) described the dependence of the work hardening moduli on strain gradients with the help of dislocation density tensor. [Regueiro et al. \(2002\)](#) derived a relation between dislocation density tensor and back stresses. [Shizawa and Zbib, 1999](#), [Menzel and Steinmann, 2000](#), [Gurtin, 2002](#) also used the dislocation density tensor with kinematic hardening which incorporated second order strain gradients. [Yefimov and Van der Giessen, 2005](#), [Han et al., 2005a](#), [Han et al., 2005b](#), [Cheong et al., 2005](#) incorporated the geometrically necessary dislocation densities with in the slip system resistance. [Ohashi \(2005\)](#) suggested a model where the geometrically necessary dislocations densities are incorporated in the strain hardening moduli while like other strain gradient crystal plasticity theories geometrically necessary dislocation densities are obtained from spatial gradients of the plastic shear strain on slip systems. [Brinckmann et al. \(2006\)](#) proposed a first-order strain gradient model based on dislocation density. The results are compared with the mechanism-based strain gradient plasticity theory of [Huang et al. \(2004\)](#) and are found to be consistent. [Al-Rub and Voyiadjis \(2006\)](#) proposed a physically motivated mathematical model of strain gradient plasticity. The dislocation processes which include generation, motion, immobilization, recovery, and annihilation are included in statistically stored dislocations while geometrically necessary dislocations are defined as a function of the gradient of plastic strain (shear) gradient.

Strain gradient theories are commonly classified as higher- and lower-order theories, higher-order theories include higher-order stress which increase the order of governing balance equations and also additional boundary conditions are needed (for example, [Shu and Fleck, 1998](#), [Shu et al., 1999](#), [Gurtin, 2002](#), [Al-Rub and Voyiadjis, 2005](#)). In lower-order theories neither additional higher-order stress quantities are involved nor additional boundary conditions ([Acharya and Bassani, 2000](#), [Meisssonier et al., 2001](#), [Yefimov and Van der Giessen, 2005](#), [Han et al., 2005a](#), [Han et al., 2005b](#), [Cheong et al., 2005](#), [Ohashi, 2005](#)). Therefore, the lower-order strain gradient theories preserve the classical structure of conventional crystal plasticity (as discussed in Section 2) and also it does not require higher-order stress or additional boundary conditions, on the other hand, these lower-order strain

cannot capture, for example, the formation of boundary layers in case of hindered plasticity ([Shu and Fleck,](#)

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1999, Gurtin, 2002). The numerical implementation of higher-order theories with additional boundary conditions and higher-order governing balance equations have been discussed in detail in Shu et al., 1999, Al-Rub and Voyiadjis, 2005.

The mechanism-based strain gradient theory (MSG) which is being used here is developed in Han et al., 2005a, Han et al., 2005b. It is a generalization of the conventional crystal plasticity theory (Hill, 1967, Rice, 1971, Hill and Rice, 1972) and as discussed above is a lower-order theory. The mechanism-based strain gradient crystal plasticity theory reduces to conventional crystal plasticity theory when strain gradients vanish or when the scale of the non-uniform deformation is larger than the intrinsic length scale l . This will be shown in the following. The intrinsic length scale is given by (Han et al., 2005a, Han et al., 2005b):

$$l = \frac{\alpha^2 \mu^2 b}{(g_0)^2} \quad (8)$$

where α is an empirical coefficient ranging from 0.3 to 0.5, μ is the shear modulus, b the magnitude of Burgers vector while g_0 denotes a reference slip resistance which is taken to be $\mu/100$ (Han et al., 2005a).

In mechanism-based strain gradient crystal plasticity theory (Han et al., 2005a, Han et al., 2005b), the effective slip resistance g_T^α is the resultant of the slip resistance due to strain hardening which is caused by statistically stored dislocations g_S^α for slip system α and the slip resistance due to the effective density of geometrically necessary dislocations g_G^α .

$$g_T^\alpha = \sqrt{(g_S^\alpha)^2 + (g_G^\alpha)^2} \quad (9)$$

The slip resistance due to strain hardening which is caused by statistically stored dislocations g_S^α is computed using equation, given as

$$\dot{g}_s^{(\alpha)} = \sum_{\beta} h_{\alpha\beta} \dot{\gamma}^{(\beta)}$$

The intrinsic length scale l enters into the effective slip resistance g_T^α through the slip resistance contribution due to the effective density of geometrically necessary dislocations η_G^α .

$$\eta_G^\alpha = \left| \frac{\partial \gamma^\alpha}{\partial x} \right|$$

which is given by [Han et al. \(2005a\)](#)

$$\eta_G^\alpha = |\underline{m}^\alpha \times \sum_\beta s^{\alpha\beta} \nabla \gamma^\beta \times \underline{m}^\beta| \quad (10)$$

where $\underline{m}^\alpha$ is the slip plane normal while $s^{\alpha\beta} = \underline{s}^\alpha \cdot \underline{s}^\beta$ with $\underline{s}^\alpha$ is the slip direction and $\times$ denotes the vector product.

The slip resistance due to the effective density of geometrically necessary dislocations g_G^α is given by

$$g_G^\alpha = \sqrt{l \eta_G^\alpha} \quad (11)$$

The modified effective slip resistance is, therefore, given by

$$g_T^\alpha = \sqrt{(g_S^\alpha)^2 + l \eta_G^\alpha} \quad (12)$$

Using the above equation of effective slip resistance, the plastic slip rate is then modified to

$$\dot{\gamma}^\alpha = \dot{\gamma}_0^\alpha \operatorname{sgn}(\tau^{(\alpha)}) \left\{ \left| \frac{\tau^{(\alpha)}}{g_T^{(\alpha)}} \right| \right\}^m \quad (13)$$

In what follows next, the above discussed relations for mechanism-based strain gradient crystal plasticity theory have been used to study strain gradient effect on the stresses around an interface crack tip along with the J -integral values. As the mechanism-based strain gradient crystal plasticity theory is a lower-order theory, therefore, it neither involves any higher-order stress nor additional boundary conditions beyond those in conventional crystal plasticity theory. In other words, all governing equations are identical to those in conventional crystal plasticity theory except that the incremental plasticity moduli (see Eq. (12)) are changed to account for the strain gradient effects ([Niordson and Hutchinson, 2003](#)).

The UMAT subroutine consisting of conventional crystal plasticity theory [Huang \(1991\)](#) has been modified (the implementation of conventional crystal plasticity theory in the finite element code, ABAQUS, has been discussed in detail in [Huang, 1991](#), [Kysar, 2001a](#), [Kysar, 2001b](#)) and above discussed relations of mechanism-based crystal plasticity theory (Eqs. (11), (12), (13), (14), (15), (16)) have been implemented in the UMAT subroutine in order to study the effect of strain gradient theory on the stresses around the crack tip and fracture energies. The implementation is straightforward, only few modifications are made in the conventional

subroutine ([Huang, 1991](#)). The intrinsic length scale l has been computed using Eq. (11), $l = \frac{\alpha^2 \mu^2 b}{(g_0)^2}$. The effective

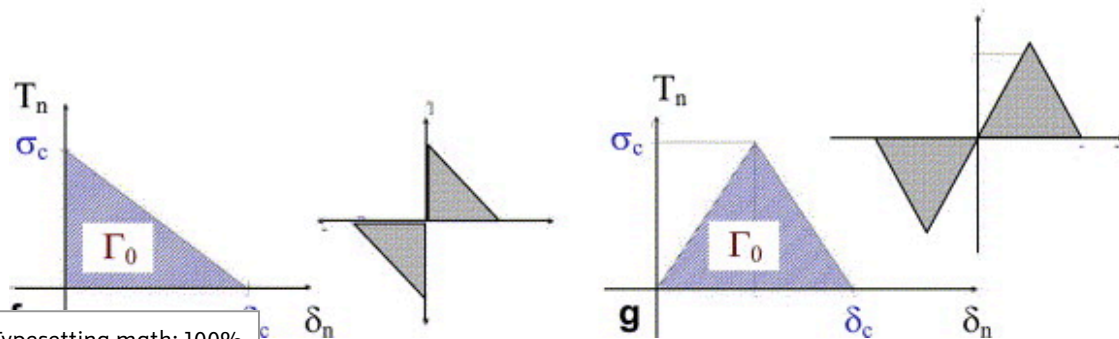
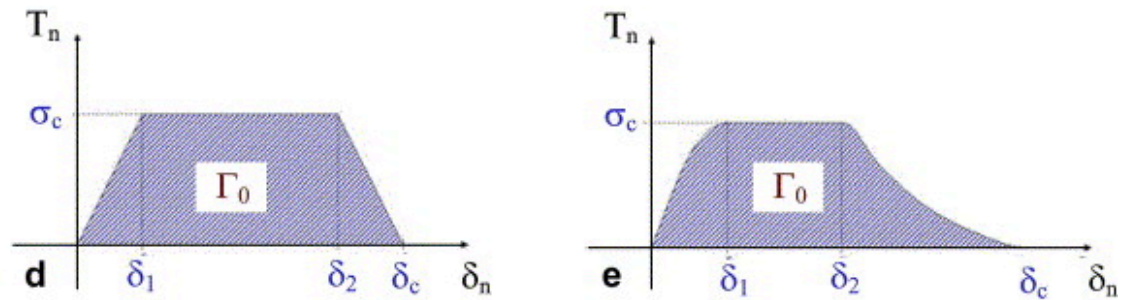
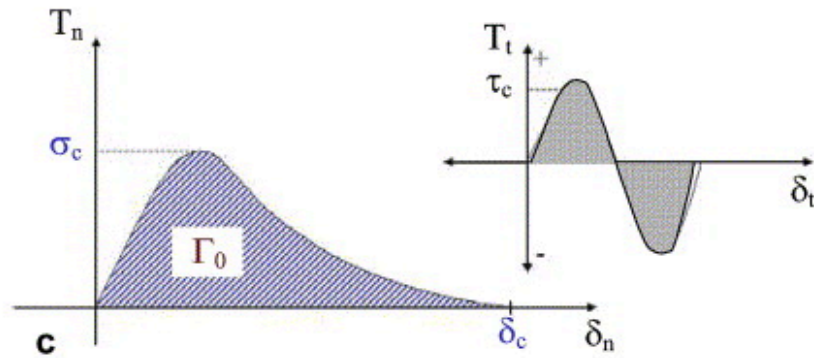
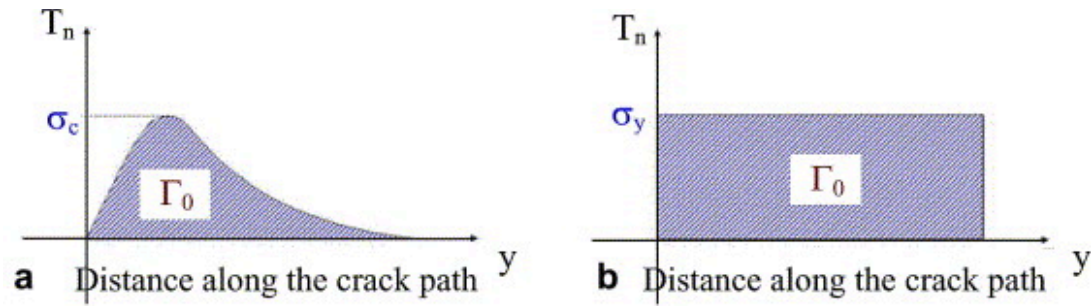
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density of geometrically necessary dislocations η_G^α is computed using Eq. (13), $\eta_G^\alpha = |\underline{m}^\alpha \times \sum_\beta s^{\alpha\beta} \nabla \gamma^\beta \times \underline{m}^\beta|$. The slip resistance due to the effective density of geometrically necessary dislocations g_G^α is computed using Eq. (14), $g_G^\alpha = \sqrt{l \eta_G^\alpha}$. Finally the modified effective slip resistance is obtained using Eq. (15), $g_T^\alpha = \sqrt{(g_S^\alpha)^2 + l \eta_G^\alpha}$. This modified effective slip resistance is then used in the viscoplastic law $\dot{\gamma}^\alpha = \dot{\gamma}_0^\alpha \text{sgn}(\tau^{(\alpha)}) \left\{ \left| \frac{\tau^{(\alpha)}}{g_T^{(\alpha)}} \right| \right\}^m$ already implemented in Huang (1991).

4. The cohesive model

Cohesive models have found increasing interest to simulate fracture in metallic, polymeric, and ceramic materials and their composites (Barenblatt, 1959, Dugdale, 1960, Xu and Needleman, 1994, Tvergaard and Hutchinson, 1992, Camacho and Ortiz, 1996, Geubelle and Baylor, 1998, Scheider and Brocks, 2003). Using cohesive models, the fracture can be simulated for any structure with or without a crack. The idea for the cohesive model is based on the consideration that the fracture mechanics analysis presupposes the existence of an infinitely sharp crack leading to singular crack tip fields. However, in real materials neither the sharpness of the crack nor the stress levels near the crack tip region can be infinite. Barenblatt, 1959, Dugdale, 1960 were the first ones to propose the concept of cohesive model to overcome these difficulties.

Barenblatt (1959), who investigated the fracture of brittle materials, defined the traction along the crack path as a function of the crack tip distance along the crack front. The typical traction-crack tip distance is shown in Fig. 1(a).



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Fig. 1. Various cohesive laws proposed by different authors: (a) [Barenblatt \(1959\)](#); (b) [Dugdale \(1960\)](#); (c) [Xu and Needleman \(1994\)](#); (d) [Tvergaard and Hutchinson \(1992\)](#); (e) [Scheider and Brocks \(2003\)](#); (f) [Camacho and Ortiz \(1996\)](#); (g) [Geubelle and Baylor \(1998\)](#).

[Dugdale \(1960\)](#) introduced the finite stress at the crack tip to be the yield stress, which is not true always as the crack-opening stress can be much higher than the yield stress. The typical curve is shown in [Fig. 1\(b\)](#).

[Xu and Needleman \(1994\)](#) was the one of the first to use polynomial and exponential types of traction–separation equations to simulate the particle debonding in metal matrices. The exponential fit is used for normal traction while trigonometric fit for shear traction. The traction–separation curves are plotted in [Fig. 1\(c\)](#). [Xu and Needleman \(1994\)](#) further used the above models to study the void nucleation at the interface of particle and matrix metal. [Tvergaard and Hutchinson \(1992\)](#) used a trapezoidal shape of the traction–separation model to calculate the crack growth resistance in elasto-plastic materials. The typical curve is shown in [Fig. 1\(d\)](#). [Tvergaard \(1990\)](#) also extended the [Needleman \(1987\)](#) model of pure normal separation for mixed mode loading. This is one of the most popular cohesive laws used by many authors (e.g., [Chaboche et al., 1997](#)). The typical curve is shown in [Fig. 1\(d\)](#).

[Camacho and Ortiz \(1996\)](#) employed a linear cohesive fracture mode to propagate multiple cracks along arbitrary paths during impact damage in brittle materials. This model predicts failure by both shear and normal separation in tension and by shear separation in compression. The typical curve is plotted in [Fig. 1\(f\)](#).

[Geubelle and Baylor \(1998\)](#) utilized a bilinear cohesive model to simulate spontaneous initiation and propagation of cracks in thin composite plates subjected to low-velocity impact. The traction separation curves for this model are shown in [Fig. 1\(g\)](#).

A cohesive law proposed by [Scheider and Brocks \(2003\)](#) has been used for the case of a growing crack. Cohesive models have found increasing interest to simulate fracture in metallic, polymeric, and ceramic materials and their composites. Using cohesive models, the fracture can be simulated for any structure with or without an initial crack. The idea for a cohesive model is based on the consideration that the fracture mechanics analysis presupposes the existence of an infinitely sharp crack leading to the singular crack tip fields. However, in real materials neither the sharpness of the crack nor the stress levels near the crack tip are generally infinite.

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In the cohesive law proposed by [Scheider and Brocks \(2003\)](#) the initial stiffness of the cohesive element can be varied by changing δ_1 , as shown in [Fig. 2](#). If δ_1 is increased the slope of the curve before σ_c is reached, decreases and similarly the slope of the curve increases if δ_1 is decreased. Also a region can be defined, where the traction in the cohesive element is kept constant. This has been achieved by using two additional parameters δ_1 and δ_2 ([Fig. 2](#)), leading to the following formulation for the function $\sigma(\delta)$:

$$\sigma(\delta) = \sigma_c \begin{cases} 2\left(\frac{\delta}{\delta_1}\right) - \left(\frac{\delta}{\delta_1}\right)^3 & \delta \leq \delta_1 \\ 1 & \delta_1 \leq \delta \leq \delta_2 \\ 2\left(\frac{\delta - \delta_2}{\delta_c - \delta_2}\right)^3 - 3\left(\frac{\delta - \delta_2}{\delta_c - \delta_2}\right)^2 + 1 & \delta_2 \leq \delta \leq \delta_c \end{cases} \quad (14)$$

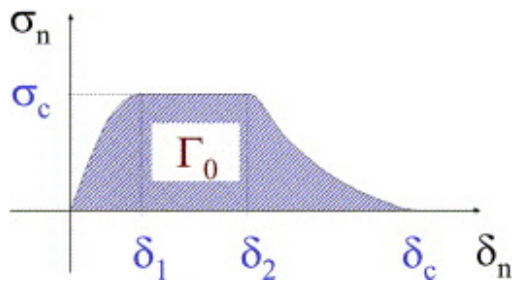
Both normal and shear behaviour are described by the same relation as given in Eq. (14). In order to take into account the mode mixity effects an additional normal and shear coupling function is introduced (see Eq. (15)), this function reduces the maximum traction as tangential separation increases.

$$g(\delta) = 2\left(\frac{\delta}{\delta_0}\right)^3 - 3\left(\frac{\delta}{\delta_0}\right)^2 + 1 \quad (15)$$

With normal and shear coupling term (Eq. (15)), the mixed mode traction separation law is then given by

$$\sigma_{m \text{ mod } e}(\delta) = \sigma(\delta)g(\delta) \quad (16)$$

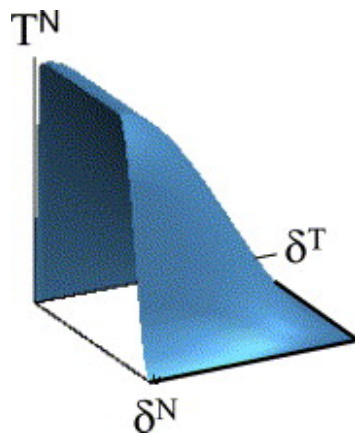
where $\sigma_{m \text{ mode}}(\delta)$ is the traction with mixed mode coupling term. The surface plot of Eq. (16) is shown for normal traction in [Fig. 3](#). It can be seen as tangential separation δ_T increases the traction value decreases for the same value of normal separation δ_n .



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Fig. 2. Cohesive law proposed by [Scheider and Brocks \(2003\)](#).

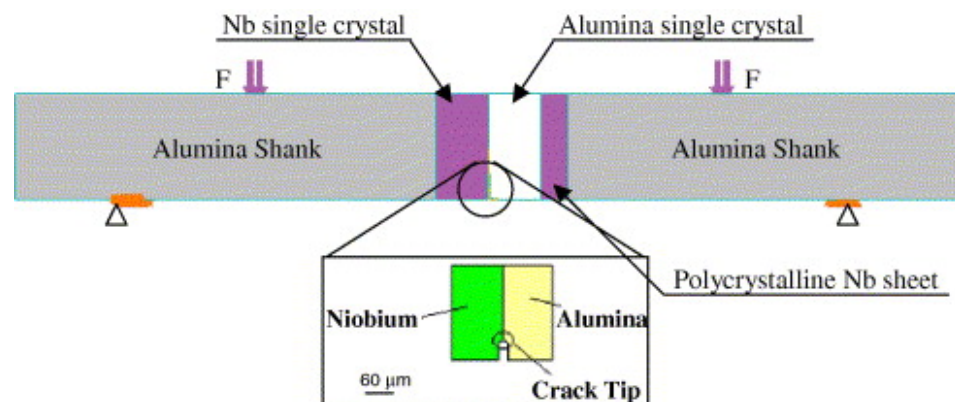


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Fig. 3. Mixed mode traction–separation law ([Scheider and Brocks, 2003](#)).

5. Finite element model

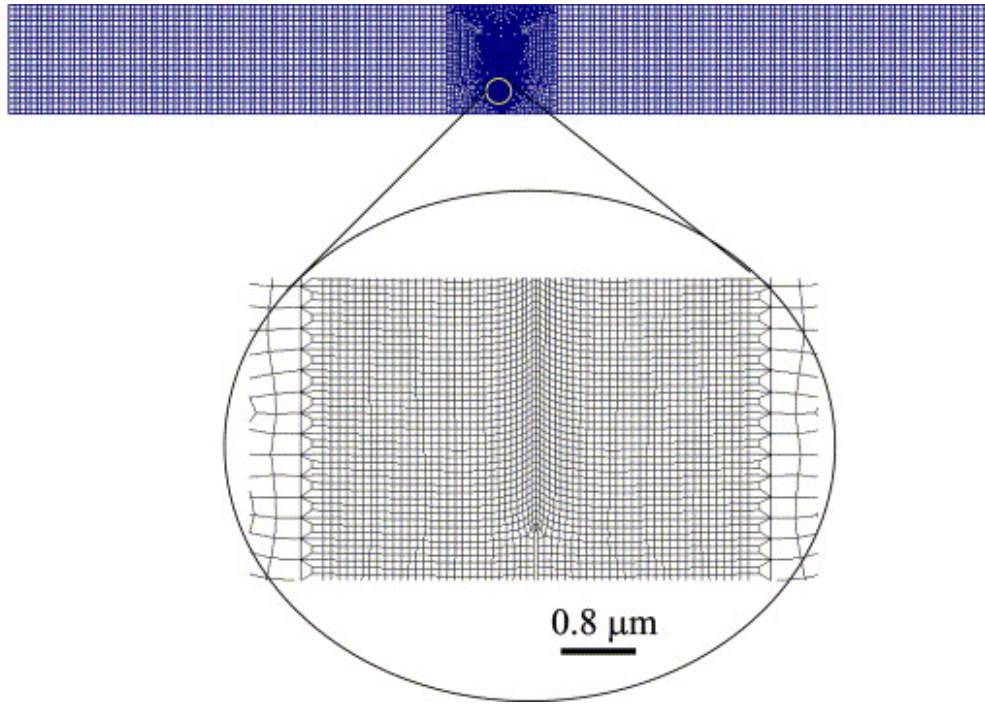
The finite element model is based on the experiments performed in [Korn, 1993](#), [Korn et al., 2002](#), in which the influence of orientation and impurities on the fracture behaviour of Nb/sapphire interfaces were studied using notched bending tests. The specimens to perform four-point bending test experiments in [Korn, 1993](#), [Korn et al., 2002](#) were prepared by diffusion bonding single crystals of niobium and alumina in a ultra high vacuum furnace for different interface orientations. It was found that for undoped bicrystals bonded at 1400°C, the computed interfacial fracture energy ranged from 77 to 2100J/m² depending on the interface planes of Nb and sapphire. For fracture evaluation, interfacially notched bending test specimens of dimensions 2×4×36mm³ were prepared. The specimen is shown in [Fig. 4](#). The notch length is 0.4mm. The specimen is loaded to the fracture load F_c in a four-point bending tests device at a crosshead speed of 96.8μm/min. The load F and the crosshead deflection at the load points v are simultaneously recorded.



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Fig. 4. Four-point bending test specimen.

The finite element model is constructed based on the information provided in [Korn, 1993](#), [Korn et al., 2002](#). As shown in [Fig. 4](#), for the case of stationary crack tip the finite element model of niobium single crystal is bonded to the alumina using bond option available in ABAQUS while for the case of growing crack the bond option at the interface was removed with a layer of cohesive elements of zero thickness. The cohesive law used for this study is proposed by [Scheider and Brocks \(2003\)](#). Finite element models of alumina shanks and polycrystalline niobium sheet was always joined with the corresponding niobium single crystal finite element model and alumina finite element model using tie constraint available in ABAQUS. The finite element mesh consisted of 39,508 plane strain four-noded quadrilateral elements, as shown in [Fig. 5](#). As the mechanism-based strain gradient crystal plasticity theory is a lower-order theory, therefore linear elements can be used because strain gradient never appears directly as an independent variable. While the use of linear elements for the higher-order theory causes problems due to the fact that in higher-order theory not only strain, but also strain gradient play as independent variables, therefore if linear displacement element are used for higher-order theory then the strain is constant and strain gradient always vanishes. In other words, mechanism-based strain gradient crystal plasticity theory is a lower-order theory which does not involve the strain gradient directly. Only the gradient of plastic strain (not total strain) is evaluated via the interpolation from the Gaussian integration points ([Huang et al., 2004](#)). Throughout this work implicit integration scheme is used.



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Fig. 5. Finite Element Mesh of four-point bending test bicrystal specimen.

For all simulations, both outer alumina shanks (ceramic) and the alumina single crystal at the middle of the specimen were treated as purely elastic with a Young's modulus of 390GPa and a Poisson's ratio of 0.27. The polycrystalline niobium sheet is always modelled with an elastic–plastic constitutive law. Young's modulus and Poisson's ratio for the polycrystalline niobium sheet were the same for all simulations ($E=104.9\text{GPa}$, $\nu=0.397$). These elastic and plastic data are adjusted to alumina and niobium, respectively, given in [Kohnle et al., 2000](#), [Kohnle et al., 2002](#). The plastic behaviour of the stress–strain curve of the polycrystalline niobium sheet is approximated by a Ramberg–Osgood relation ([Kohnle et al., 2000](#)), which is described in a one-dimensional case by the following equation:

$$E_c \varepsilon = \sigma + \alpha \left(\frac{\sigma}{\sigma_0} \right)^{n-1} \sigma \quad (17)$$

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here, n denotes the hardening exponent, α the yield offset and σ_0 the yield stress. This material law is nonlinear from the

beginning, but for commonly used hardening exponents ($n \geq 5$) the divergence from linearity is only slight for stresses below σ_0 . The chosen plasticity theory is the deformation plasticity theory (for details see [Kohnle et al., 2000](#)) and references therein, which describes not a plastic material behaviour, but a nonlinear elastic material. This means, that no unloading criterion exists. This choice of the theory to model polycrystalline niobium sheet has no effect on the current analyses as during all the simulations it is found that niobium polycrystalline sheet is always in the elastic region. The parameters of the above equation are adjusted to the niobium stress–strain curves in ([Kohnle et al., 2000](#), [Kohnle et al., 2002](#)). The parameter used are $n=6$, $\sigma_0=180\text{MPa}$ and $\alpha=0.3$. As mentioned before single crystal niobium is modelled using the hardening law of [Bassani and Wu, 1991](#), [Bassani and Wu, 1993](#). The hardening parameters for each slip system are derived in [Siddiq and Schmauder \(2005a\)](#). The hardening parameters used for each family of slip systems, i.e., $(110)[111]$ and $(112)[111]$, are given in [Table 1](#). Slip systems $(123)[111]$ were not considered during the simulation, this is based on the experimental findings of [Bowen et al., 1967](#), [Ma and Raabe, 2004](#), his experimental results showed that $(123)[111]$ slip systems are only activated when substitutional impurities are present in the single crystalline niobium.

Table 1. Hardening parameters for Bassani & Wu hardening law

	τ_0 (MPa)	τ_s (MPa)	h_0 (MPa)	h_s (MPa)	γ_0	γ_{0I}	$f_{\alpha\beta}$	$f_{\alpha\beta I}$	q	q_1
$(110)[111]$	13.7	16.4	292.262	1.4	0.25	0.25	10.0	9.9	0.2891	0.2315
$(112)[111]$	13.07	16.344	49.03325	39.2266	0.1	0.1	0.34	0.3	0.01	0.011

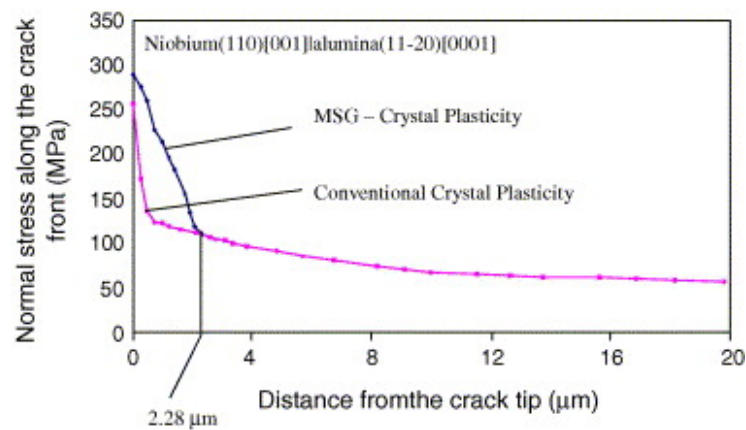
For the case of mechanism-based strain gradient crystal plasticity, the values of the coefficients α , and b used in defining the intrinsic material length parameter l (Eq. (11)) are taken from [Qu et al. \(2004\)](#) with the magnitudes of Burgers vector $b=0.25\text{nm}$, coefficient $\alpha=0.5$. These parameters lead to the value of the intrinsic material length for angle crystalline niobium to be, $l=5.29\mu\text{m}$ while for the case of conventional crystal plasticity theory the intrinsic material length (l) was taken to be zero. All the simulations have been performed for the niobium $(110)[001]$ ||alumina $(11-20)[0001]$ -interface.

6. Results and discussion

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6.1. Stationary crack tip

The distribution of normal stress along the bicrystal niobium/alumina interface ahead of the crack tip is plotted in Fig. 6 for the case of stationary crack tip. The plot shows the results of both, conventional crystal plasticity and mechanism-based strain gradient crystal plasticity theory.



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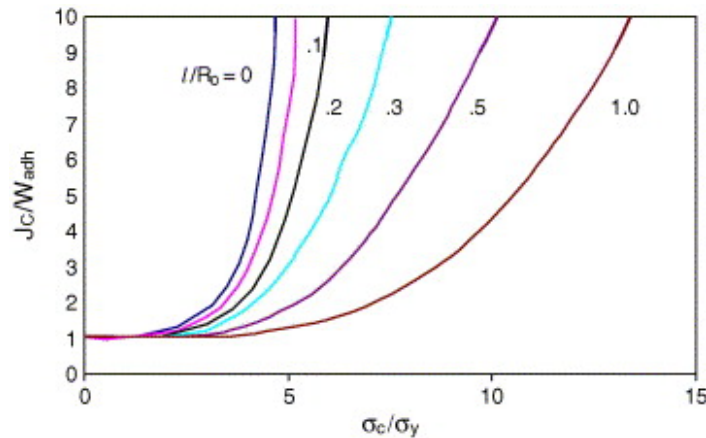
Fig. 6. Normal stress distribution along the crack front for the case of conventional crystal plasticity and MSG – crystal plasticity.

Both conventional crystal plasticity theory and mechanism-based strain gradient crystal plasticity theory give the same stress distribution at a distance of more than 2.28 μm from the crack tip. For the distance less than 2.28 μm from the crack tip, mechanism-based strain gradient crystal plasticity theory gives higher normal stress than conventional crystal plasticity. This is due to the strain gradient effect at the metal/ceramic interface. If we compare the ratio of the normal stress at the crack tip to yield stress of single crystalline niobium (σ/σ_y) then for conventional crystal plasticity theory this ratio is 5.0953 while for mechanism-based strain gradient crystal plasticity theory this ratio is increased to 5.7075 (with $\sigma_y=50.63$ MPa for niobium (110) [001]).

This suggests that a strain gradient does increase the normal stress near the crack tip significantly, but the difference apart from the crack tip is of the factor larger than 2. This difference (factor larger than 2) in stress can be explained on the basis of results obtained in [Wei and Hutchinson \(1998\)](#) where the effect of J_2 strain gradient plasticity theory on the fracture energy of the bimaterial interface is discussed in [Wei and Hutchinson \(1998\)](#). Fig. 7 shows the ratio of fracture energy to work of adhesion

Typesetting math: 100% section of cohesive strength to yield stress ratio (σ_c/σ_y) for various values of l/R_0 for a homogeneous material,

where R_0 is the size of the plastic zone. It can be seen that at a given (J_c/W_{adh}) value, (σ_c/σ_y) increases as l/R_0 increases and that at a given σ_c/σ_y ratio J_c/W_{adh} increases as l/R_0 decreases. These results show that l/R_0 has a strong influence on the level of stress that will be achieved around the crack tip due to strain gradient effects, i.e., the higher the l/R_0 value the higher will be the stresses around the crack tip.



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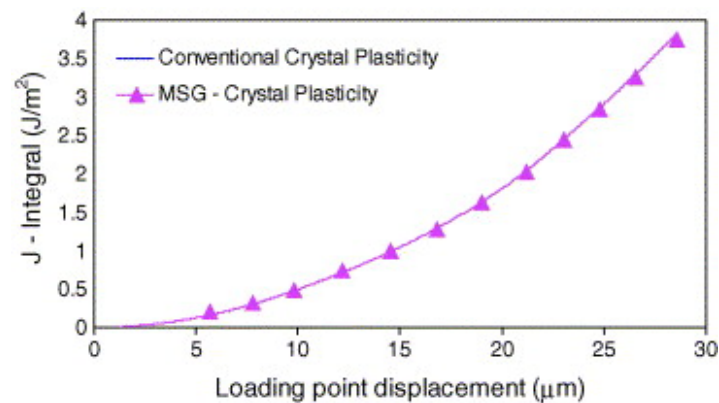
Fig. 7. The influence of strain gradient plasticity on (J_c/W_{adh}) for homogeneous material (Wei and Hutchinson, 1998, for details see text).

If we now look into the current niobium/alumina bicrystal interface the value of intrinsic material length is $l=5.29\mu\text{m}$ as computed above. Also, the size of the plastic zone at this stress level is found to be $32.5\mu\text{m}$. The size of plastic zone is an averaged size around the crack tip which is obtained by taking the mean of all the distances around the crack tip until which the plastic deformation occurs. The size of plastic zone is also obtained by using the relation proposed by Tvergaard and Hutchinson (1993) (for details see references therein) which also gives a similar value for the current niobium/alumina bicrystal interface $R_0\sim 32\mu\text{m}$. This gives a value of $l/R_0=0.1628$. This is the region where (σ_c/σ_y) is less sensitive to the l/R_0 value due to the reason that the scale of plastic deformation is larger than the intrinsic material length l and that is why mechanism-based strain gradient crystal plasticity theory has only a small influence on the (σ_c/σ_y) ratio for the case of niobium/alumina bicrystal interfaces for all interface orientations. It should also be noted that if a maximum stress criterion is used as fracture criterion rather than the

where a critical stress value at a distance ahead of the crack tip is prescribed then the difference in maximum

stress obtained from both conventional and mechanism-based crystal plasticity theory will be sensitive to the distance prescribed ahead of the crack tip. For the case of niobium/alumina interface the region is found to be in the range of 0.0–1.9 μm .

The fracture energies computed for the case of mechanism-based strain gradient crystal plasticity theory and conventional crystal plasticity theory were found to be identical, as shown in Fig. 8. This is due to the limited region (0.1–2 μm) where stresses were found to be higher for the case of mechanism-based strain gradient crystal plasticity theory.

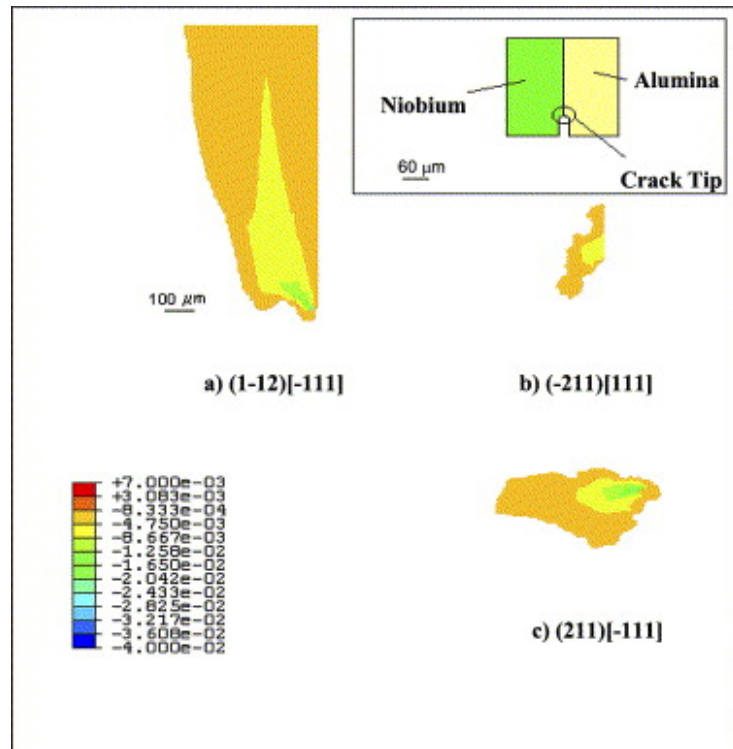


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Fig. 8. J -integral vs loading point displacement for conventional crystal plasticity and MSG-crystal plasticity.

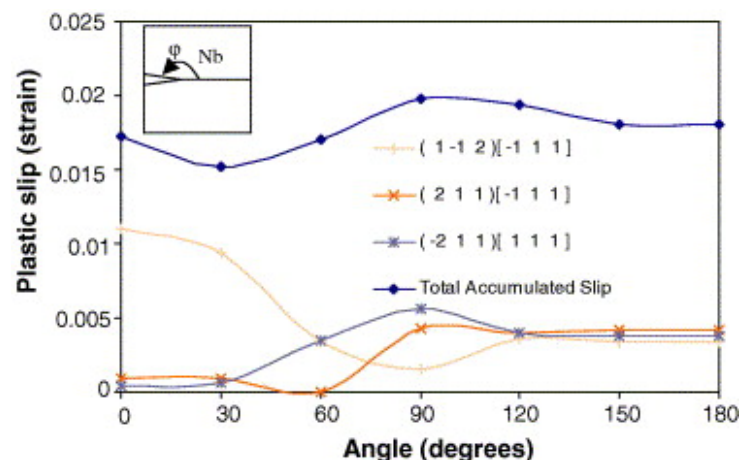
It is also found that for niobium (110)[001] orientation (1–12)[–111], (–211)[111] and (211)[–111] systems play a major role in the total plastic slip and eventually in the plastic strain (see Fig. 9). The reason is the highest Schmidt factor for these (1–12)[–111], (–211)[111] and (211)[–111] systems, i.e. ~ -0.4714 , -0.2357 and -0.408 while other slip systems have smaller Schmidt factor ~ -0.2 or even less. Fig. 10 shows the relative magnitudes of the slip activity, where the slip activity at a radius of 15 μm (it is a reasonable distance from the crack tip where influence of all the activated slip system was seen during the simulations; distances other than 15 μm give similar results, i.e. same slip systems contributing in total accumulated slip but only the magnitude of plastic strain changes) from the crack tip is plotted for each of the activated slip systems relative to the polar angle ϕ ; 0° coincides with the prolongation of the crack and 180° coincides with the single crystal niobium crack flank. It can be easily identified which slip system is contributing the highest amount of plastic slip in different regions around the crack tip. Fig. 10 shows that the maximum contribution of plastic slip comes from the slip system (1–12)[–111] for the polar angle ϕ range of 0° – 60° while the contribution of slip systems (–211)[111] and (211)[–111] is maximum for the range of 60° – 120° .

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Fig. 9. Plastic strain on various slip systems at the tip of a stationary crack for orientation II.



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Fig. 10. Plastic strain on slip systems at $15\mu\text{m}$ radius from the tip of a stationary crack for orientation II.

6.2. Quasistatically growing crack

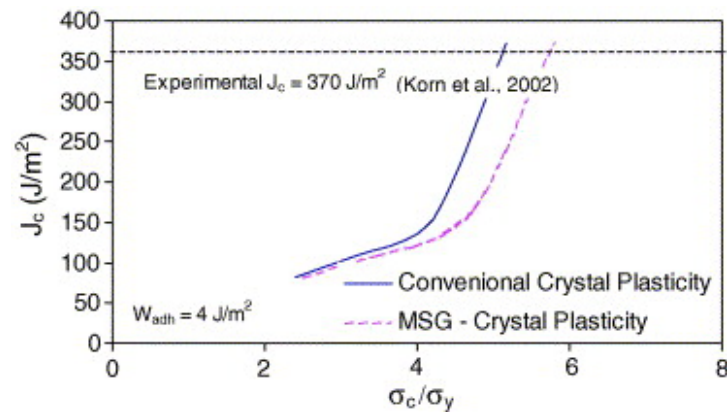
All finite element analyses of crack growth have been performed for the case when niobium $(110)[001]$ is bonded with alumina $(11-20)[0001]$ using crystal plasticity theory where single crystalline niobium is always modelled using the Bassani & Wu hardening law. The experimental fracture energy (Korn et al., 2002) for this orientation was found to be 370J/m^2 for a bonding temperature of 1400°C .

Parametric studies have been performed to study the effect of cohesive law parameters, such as cohesive strength and cohesive energy (also known as work of adhesion). This is being done for calibrating the cohesive law parameters, in order to reach the experimentally found fracture energy of this metal/ceramic system. Both conventional crystal plasticity theory and mechanism-based strain gradient crystal plasticity theory will be used for the analyses.

The effect of cohesive strength on the fracture energy of bicrystal niobium $(110)[001]$ ||alumina $(11-20)[0001]$ has been studied by keeping the work of adhesion to be 4.0J/m^2 . Fracture energy for various values of cohesive strength/yield stress ratios is plotted in Fig. 11. These results depict that as the cohesive strength increases, the fracture energy also increases gradually until

Typesetting math: 100% s value of σ_c/σ_y the slope changes and the fracture energy becomes more sensitive to cohesive strength, this is due to the higher plastic dissipation as cohesive strength increases. Tile results also show that the σ_c/σ_y ratio required for

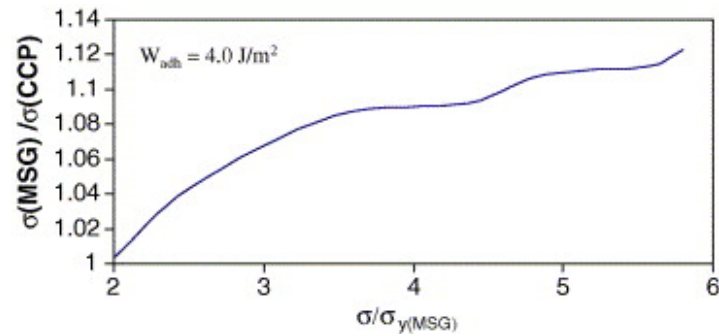
mechanism-based crystal plasticity theory is 5.7 while the σ_c/σ_y value obtained using conventional crystal plasticity theory is 5.1 for the same macroscopic fracture energy value of 370J/m².



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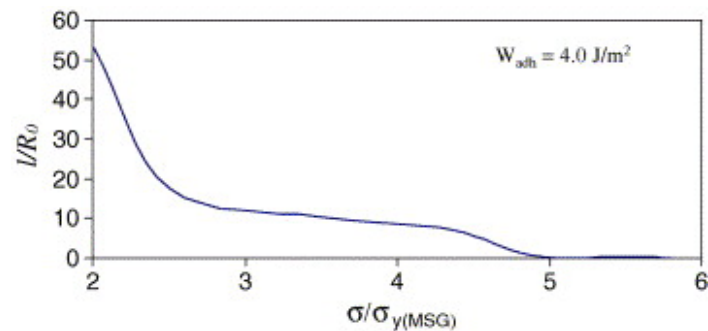
Fig. 11. Fracture energy obtained from conventional and mechanism-based strain gradient crystal plasticity theory for different cohesive strength/yield stress ratios.

From the same set of analyses, cohesive strength ratio ($\sigma_{(MSG)}/\sigma_{(CCP)}$) is plotted as function of $(\sigma/\sigma_y)_{(MSG)}$ in Fig. 12 for the work of adhesion value of 4J/m², where $CT_{(MSG)}$ denotes the cohesive strength obtained from mechanism-based strain gradient crystal plasticity theory while $\sigma_{(CCP)}$ is the cohesive strength obtained from conventional crystal plasticity theory (cohesive strength, in general, is the peak stress level attained by the cohesive elements during deformation at which damage initiates). It shows that as $(\sigma/\sigma_y)_{(MSG)}$ increases, the value of $(\sigma_{(MSG)}/\sigma_{y(CC)})$ also increases and then starts to saturate after a $(\sigma/\sigma_y)_{(MSG)}$ value of 3.5. This is due to the decrease in l/R_0 ratio as shown in Fig. 13, which decreases as the cohesive strength increases due to the higher plastic dissipation and larger plastic zone.



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Fig. 12. Ratio of the cohesive strength obtained using MSG-crystal plasticity theory and conventional crystal plasticity theory ($\sigma_{\text{MSG}}/\sigma_{\text{y(CCP)}}$) for different ratios of stress normal to the crack tip to the yield stress obtained from MSG-crystal plasticity theory $(\sigma/\sigma_{\text{y}})_{\text{(MSG)}}$.



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Fig. 13. Ratio of the intrinsic material length to size of plastic zone (l/R_0) for different ratios of stress normal to the crack tip to the yield stress obtained from MSG-crystal plasticity theory $(\sigma/\sigma_{\text{y}})_{\text{(MSG)}}$.

A similar study has been done by keeping the cohesive strength constant and by varying the work of adhesion from 1 J/m^2 to 9.8 J/m^2 , this change of work of adhesion does not show any influence on the size of the plastic zone and the l/R_0 ratio is always constant. The macroscopic fracture energy increases linearly with increasing work of adhesion providing the cohesive strength to be constant.

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Table 2 shows the final results of the cohesive strength, corresponding values of work of adhesion, i.e. 1.0, 4.0, and 9.8 J/m², size of plastic zone obtained from conventional and mechanism-based strain gradient crystal plasticity theory for a macroscopic fracture energy of 370 J/m² (Korn et al., 2002). It was found that as the l/R_0 ratio decreases the $(\sigma_{(MSG)}/\sigma_{y(CCP)})$ value also decreases as is discussed above and also shown in Fig. 12, Fig. 13.

Table 2. Comparison of parameters obtained from conventional and mechanism-based strain gradient crystal plasticity theory

Conventional crystal plasticity		MSG-crystal plasticity		R_0 (μm)	l/R_0	$\sigma_{c(CCP)}/\sigma_{c(MSG)}$
σ_c	W_{adh}	σ_c	W_{adh}			
283	1.0	319.6	1.0	27.4	0.193	1.13
258	4.0	289	4.0	32.5	0.163	1.12
225	9.8	248	9.8	34.8	0.152	1.102

7. Conclusion

A comparative study of the tensile stresses at the interface crack tip was presented using mechanism-based strain gradient crystal plasticity theory and conventional crystal plasticity theory. Fracture analyses were performed for a stationary crack tip and for a growing crack using a cohesive modelling approach.

It is found that by using mechanism-based strain gradient theory we can predict higher stresses at the crack tip as compared to the conventional crystal plasticity theory. Also, our results show that strain gradient effects strongly depend upon the intrinsic material length to size of the plastic zone ratio (l/R_0). We have found that as the l/R_0 ratio increases the normal stresses reached using mechanism-based strain gradient crystal plasticity theory are much larger than the stresses obtained from conventional crystal plasticity theory. We also presented an insight about the role of cohesive strength and work of adhesion on fracture energies. This information will help experimentalists design better materials.

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